

Program : Diploma in Polymer Technology	
Course Code : 2091	Course Title: Fundamentals Of Polymer Science
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3,T:0,P:0)	Periods per semester: 45

Course Objectives:

- To understand the basic characteristics of carbon and organic compounds .
- To understand the history, general characteristic and classification of polymers
- To explain the chemistry of polymerization and polymerization techniques
- To generalize the polymer degradation

Course Prerequisites:

Topic	Course code	Course name	Semester
Classification of polymers		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the basic characteristics of carbon and organic compounds	10	Understanding
CO2	Describe the history, general characteristic and classification of polymers	10	Understanding
CO3	Explain the chemistry of polymerization and polymerization techniques	13	Understanding
CO4	Interpret the process of polymer degradation and its impact on environment	10	Applying
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic characteristics of carbon and organic compounds		
M1.01	Identify the uniqueness of carbon, basics and types of organic and inorganic compounds	3	Understanding
M1.02	Explain classification of organic compounds and their nomenclature	3	Understanding
M1.03	Elementary study of carbohydrates and proteins	2	Understanding
M1.04	Describe Petroleum and petrochemicals	2	Understanding
Contents: Organic chemistry Introduction - definition and scope, unique properties of carbon, difference between organic and inorganic compounds, purification of organic compounds. Hydrocarbon - Saturated and unsaturated, aliphatic and aromatic, homologues series, characteristics, Alkanes, Alkenes, Alkynes functional groups, types of functional groups, carboxylic groups and esters. Isomerism. Dienes - classification of organic compounds and their nomenclature. Preparation, properties and application of following compounds - Ethane, Ethene, Ethyne, methyl and ethyl alcohols, vinyl chemicals, styrene, benzene and toluene. Phenol and its higher homologues, amines and amides. Acetaldehyde, formaldehyde, Acetone fatty acids - stearic acid oleic acid Amino acids and proteins, Elementary study of carbohydrates Petroleum and petrochemicals - composition of petroleum, origin, refining of petroleum cracking, synthetic petrol. Important petrochemicals and their applications.			
CO2	Describe the history, general characteristic and classification of polymers		
M2.01	Describe Characteristics and application of polymers	2	Understanding

M2.02	Explain the functionality and effect of functionality in polymer formation	2	Understanding
M2.03	Explain the concept of Degree of polymerization- molecular weight of monomer and molecular weight of polymer	3	Understanding
M2.04	Illustrate the classification of polymers with example	3	Understanding
	Series Test - I	1	

Contents:

History, General characteristics and classification of polymer science History of Polymers, Characteristics and application of polymers - micromolecules, macromolecules, monomers, oligomers, polymers, repeating units - examples Functionality - Mono functionality difunctionality and poly functionality - examples. Effect of functionality in polymer formation. Degree of polymerization - molecular weight of polymers - molecular weight of monomers - simple problems. Relationship between DP and Mw. Classification of polymers - biological - non biological - Natural, synthetic and derived polymers. - Thermo plastic -thermo set polymers - Crystalline and amorphous - organic and inorganic polymers - homo chain and hetero chain polymers - homo and co - polymers - linear branched and cross linked polymers. Alternating and random co - polymers. Examples of eac

CO3	Explain the chemistry of polymerization and polymerization techniques		
M 3.01	Describe type of polymerization reaction - chain and step polymerization with examples	2	Understanding
M 3.02	Explain the steps in chain polymerization - initiation, propagation and termination.	2	Understanding
M 3.03	Describe the mechanism of free radical, ionic and coordination polymerization	5	Understanding
M 3.04	Explain different polymerization techniques	4	Understanding

Contents:

Polymerization reaction and techniques Type of polymerization reaction - chain and step polymerization with examples - steps in chain polymerization - initiation, propagation and termination. Mechanism of free radical, ionic and coordination polymerization. Living polymers, initiators and catalyst, inhibitor sand chain transfer agents with examples. Polymerisation techniques - bulk solution, suspension and emulsion polymerization - Comparison

CO4	Polymer Degradation		
M4.01	Distinguish chain and random degradation	2	Understanding
M4.02	Explain thermal degradation ,mechanical and oxidative degradation and its impact on environment	3	Applying

M4.03	Describe photo degradation - hydrolytic degradation - UV rays and Ultrasonic degradation and its impact on environment	3	Applying
M4.04	Explain the significance of anti degradants - anti oxidants and antiozonants	2	Understanding
	Series Test - II	1	
Contents: POLYMER DEGRADATION Degradation - chain and random degradation - factors of polymer degradation - Thermal degradation - stability of polymers, factors affecting thermal stability of polymers - mechanical degradation - oxidative degradation - Photo degradation -hydraulic degradation - UV rays and Ultrasonic degradation. Advantages and disadvantages of polymer degradation. Anti degradants - stabilizers - anti oxidants anti ozonant			

Text / Reference:

T/R	Book Title/Author
1	Polymer science - V.R.Gowarikar (Publisher - New Age International (P) Ltd, New Delhi- 2005 Edition
2	Text Book of Polymer Science - Billmeyer.J.R (John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition
3.	Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International (P) Ltd, New Delhi- 2005 Edition
4.	Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science

Online Resources:

Sl.No	Website Link
1	https://books.google.com
2	libguides.uml.edu/polymerscience